Problem No - 01: **Write a JAVA Program to Display Image using JFrame**

Source Code :

```java
import java.awt.FlowLayout;

import javax.swing.ImageIcon;
import javax.swing.JFrame;
import javax.swing.JLabel;

public class Image extends JFrame {
    private ImageIcon image1;
    private JLabel label1;
    private ImageIcon image2;
    private JLabel label2;

    Image(){
        setLayout(new FlowLayout());

        image1 = new ImageIcon(getClass().getResource("ice.jpg"));
        label1 = new JLabel(image1);
        add(label1);

        image2 = new ImageIcon(getClass().getResource("pust.png"));
        label2 = new JLabel(image2);
        add(label2);
    }

    public static void main(String args[]) {
        Image gui = new Image();
        gui.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        gui.setVisible(true);
        gui.pack();
        gui.setTitle("Image Program");
    }
}
```

Output:

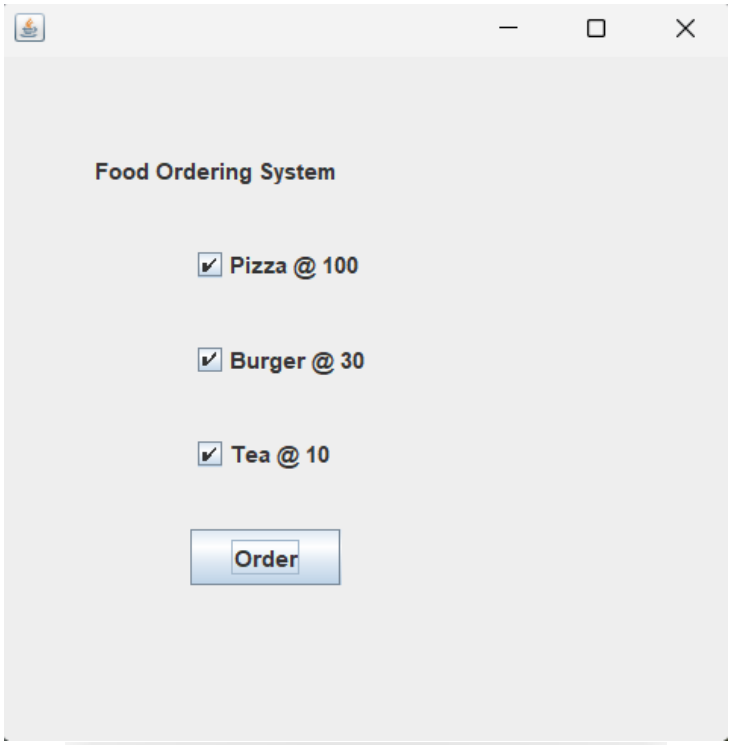
Problem No - 02: **Write a JAVA Program for generating Restaurant Bill.**

Source Code :

```
import javax.swing.*;
import java.awt.event.*;

public class BillGeneration extends JFrame implements ActionListener {
    JLabel l;
    JCheckBox cb1, cb2, cb3;
    JButton b;

    BillGeneration() {
        l = new JLabel("Food Ordering System");
        l.setBounds(50, 50, 300, 20);

        cb1 = new JCheckBox("Pizza @ 100");
        cb1.setBounds(100, 100, 150, 20);

        cb2 = new JCheckBox("Burger @ 30");
        cb2.setBounds(100, 150, 150, 20);

        cb3 = new JCheckBox("Tea @ 10");
        cb3.setBounds(100, 200, 150, 20);

        b = new JButton("Order");
        b.setBounds(100, 250, 80, 30);
        b.addActionListener(this);

        add(l);
        add(cb1);
        add(cb2);
        add(cb3);
        add(b);

        setSize(400, 400);
        setLayout(null);
        setVisible(true);
        setDefaultCloseOperation(EXIT_ON_CLOSE);
    }

    public void actionPerformed(ActionEvent e) {
        float amount = 0;
        String msg = "";

        if (cb1.isSelected()) {
            amount += 100;
            msg = "Pizza: 100\n";
        }
        if (cb2.isSelected()) {
            amount += 30;
            msg += "Burger: 30\n";
        }
        if (cb3.isSelected()) {
            amount += 10;
            msg += "Tea: 10\n";
        }

        msg += "-----------------\n";
```

```
        JOptionPane.showMessageDialog(this, msg + "Total: " + amount);
    }

    public static void main(String[] args) {
        new BillGeneration();
    }
}
```

Output:



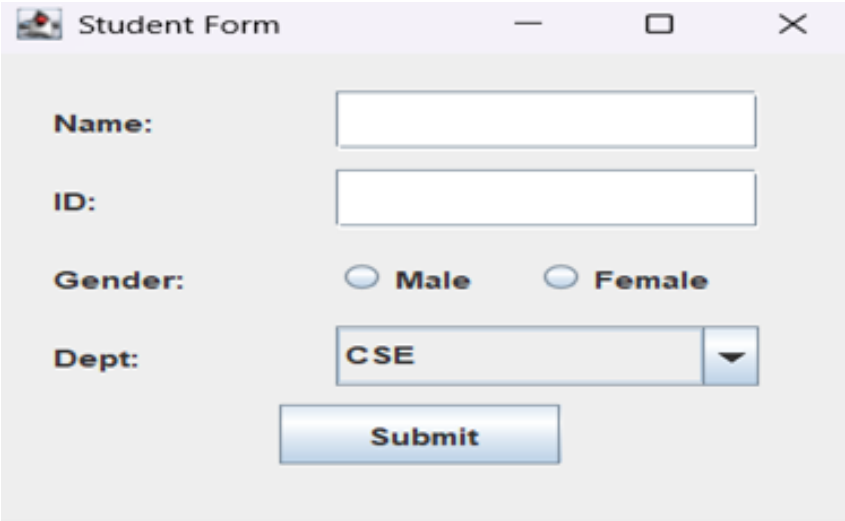
Problem No - 03: **Write a JAVA Program to Create a Student form in GUI.**

Source Code :

```
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import javax.swing.JButton;
import javax.swing.JFrame;
import javax.swing.JLabel;
import javax.swing.JPanel;
import javax.swing.JTextField;

public class form implements ActionListener {
    private static JLabel success;
    private static JFrame frame;
    private static JLabel label1, label2, label3;
    private static JPanel panel;
    private static JButton button;
    private static JTextField userText1, userText2, userText3;

    public static void main(String[] args) {
        frame = new JFrame();
        panel = new JPanel();

        frame.setSize(400, 300);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.add(panel);
        panel.setLayout(null);

        label1 = new JLabel("Name");
        label1.setBounds(10, 10, 80, 25);
        panel.add(label1);

        label2 = new JLabel("Roll");
        label2.setBounds(10, 60, 80, 25);
        panel.add(label2);

        label3 = new JLabel("Department");
        label3.setBounds(10, 110, 80, 25);
        panel.add(label3);

        userText1 = new JTextField("Enter Your Name");
        userText1.setBounds(100, 10, 200, 25);
        panel.add(userText1);

        userText2 = new JTextField("Enter Your Roll");
        userText2.setBounds(100, 60, 200, 25);
        panel.add(userText2);

        userText3 = new JTextField("Enter Your Department");
        userText3.setBounds(100, 110, 200, 25);
        panel.add(userText3);

        button = new JButton("Save");
        button.setBounds(150, 160, 80, 25);
        button.addActionListener(new form());
        panel.add(button);
```

```
        success = new JLabel("");
        success.setBounds(130, 210, 300, 25);
        panel.add(success);

        frame.setVisible(true);
    }

    @Override
    public void actionPerformed(ActionEvent e) {
        String name = userText1.getText();
        String roll = userText2.getText();
        String dept = userText3.getText();
        success.setText("Saved: " + name + ", " + roll + ", " + dept);
    }
}
```

Output:



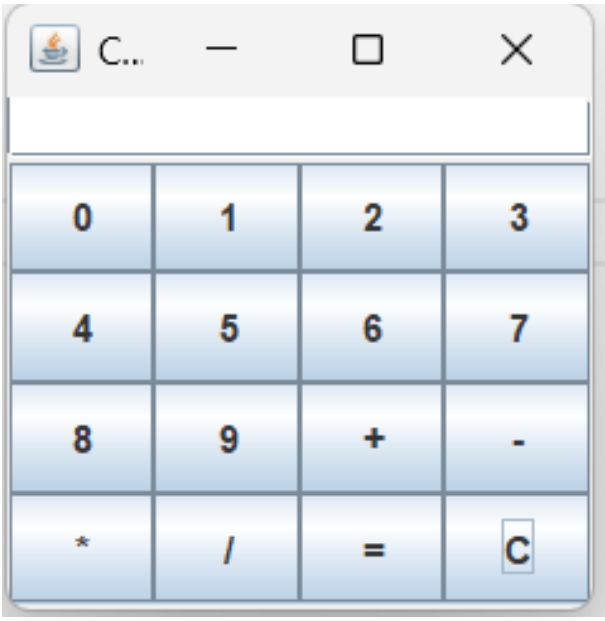
Problem No - 04: **Write a JAVA Program to develop a simple calculator in GUI.**

Source Code :

```
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import javax.swing.*;

public class calculator extends JFrame implements ActionListener {
  JButton b10, b11, b12, b13, b14, b15;
  JButton b[] = new JButton[10];
  int i, r, n1, n2;
  JTextField res;
  char op;
  public calculator() {
    super("Calulator");
    setLayout(new BorderLayout());
    JPanel p = new JPanel();
    p.setLayout(new GridLayout(4, 4));
    for (int i = 0; i <= 9; i++) {
      b[i] = new JButton(i + "");
      p.add(b[i]);
      b[i].addActionListener(this);
    }
    b10 = new JButton("+");
    p.add(b10);
    b10.addActionListener(this);
    b11 = new JButton("-");
    p.add(b11);
    b11.addActionListener(this);
    b12 = new JButton("*");
    p.add(b12);
    b12.addActionListener(this);
    b13 = new JButton("/");
    p.add(b13);
    b13.addActionListener(this);
    b14 = new JButton("=");
    p.add(b14);
    b14.addActionListener(this);
    b15 = new JButton("C");
    p.add(b15);
    b15.addActionListener(this);
    res = new JTextField(10);
    add(p, BorderLayout.CENTER);
    add(res, BorderLayout.NORTH);
    setVisible(true);
    setSize(200, 200);
  }
  public void actionPerformed(ActionEvent ae) {
    JButton pb = (JButton) ae.getSource();
    if (pb == b15) {
      r = n1 = n2 = 0;
      res.setText("");
    }
    else if (pb == b14) {
      n2 = Integer.parseInt(res.getText());
      eval();
      res.setText("" + r);
```

```
    }
    else {
      boolean opf = false;
      if (pb == b10) { op = '+'; opf = true; }
      if (pb == b11) { op = '-'; opf = true; }
      if (pb == b12) { op = '*'; opf = true; }
      if (pb == b13) { op = '/'; opf = true; }
      if (!opf) {
        for (i = 0; i < 10; i++) {
          if (pb == b[i]) {
            String t = res.getText();
            t += i;
            res.setText(t);
          }
        }
      }
      else {
        n1 = Integer.parseInt(res.getText());
        res.setText("");
      }
    }
  }
  int eval() {
    switch (op) {
      case '+': r = n1 + n2; break;
      case '-': r = n1 - n2; break;
      case '*': r = n1 * n2; break;
      case '/': r = n1 / n2; break;
    }
    return 0;
  }
  public static void main(String arg[]) {
    new calculator();
  }
}
```

Output:



Problem No - 05: **Write a Java program to illustrate suspend, resume and stop operation of thread.**

Source Code :

```
class sus_res_stop implements Runnable {
    Thread th;
    boolean suspend_flag, stop_flag;

    sus_res_stop(String name) {
        th = new Thread(this, name);
        suspend_flag = false;
        stop_flag = false;
        th.start();
    }

    public void run() {
        try {
            int j = 1;
            while (j++ < 20) {
                synchronized (this) {
                    while (suspend_flag)
                        wait();

                    if (stop_flag)
                        break;
                }
            }
        } catch (InterruptedException e) {
            System.out.println("Thread interrupted");
        }
    }

    synchronized void my_suspend() {
        suspend_flag = true;
    }

    synchronized void my_resume() {
        suspend_flag = false;
        notify();
    }

    synchronized void my_stop() {
        suspend_flag = false;
        stop_flag = true;
        notify();
    }
}

public class eg_SRS {
    public static void main(String args[]) {
        try {
            sus_res_stop S_R_S_T = new sus_res_stop("SRS");

            System.out.println("Thread S_R_S_T is created and started")
                ;
            Thread.sleep(2000);
```

```
            S_R_S_T.my_suspend();
            System.out.println("Thread S_R_S_T is suspended");
            Thread.sleep(2000);

            S_R_S_T.my_resume();
            System.out.println("Thread S_R_S_T is resumed");
            Thread.sleep(2000);

            S_R_S_T.my_stop();
            System.out.println("Thread S_R_S_T is stopped");
        } catch (InterruptedException e) {
            System.out.println("Generated interrupted exception");
        }
    }
}
```

Output:

```
Thread S_R_S_T is created and started
Thread S_R_S_T is suspended
Thread S_R_S_T is resumed
Thread S_R_S_T is stopped
```

Problem No - 06: **Write a Java program to create thread class.**

Source Code :

```java
class A extends Thread {
    public void run() {
        for (int i = 1; i <= 5; i++) {
            System.out.println("From Thread A : i = " + i);
        }
        System.out.println("Exit from A");
    }
}
class B extends Thread {
    public void run() {
        for (int j = 1; j <= 5; j++) {
            System.out.println("From Thread B : j = " + j);
        }
        System.out.println("Exit from B");
    }
}
class C extends Thread {
    public void run() {
        for (int k = 1; k <= 5; k++) {
            System.out.println("From Thread C : k = " + k);
        }
        System.out.println("Exit from C");
    }
}
class ThreadTest {
    public static void main(String args[]) {
        new A().start();
        new B().start();
        new C().start();
    }
}
```

Output:

```
From Thread A : i = 1
From Thread A : i = 2
From Thread A : i = 3
From Thread A : i = 4
From Thread A : i = 5
Exit from A

From Thread C : k = 1
From Thread C : k = 2
From Thread C : k = 3
From Thread C : k = 4
From Thread C : k = 5
Exit from C

From Thread B : j = 1
From Thread B : j = 2
From Thread B : j = 3
From Thread B : j = 4
From Thread B : j = 5
Exit from B
```

Problem No - 07: **Write a Java program to illustrate yield(), stop() and sleep() method using thread.**

Source Code :

```
class A extends Thread {
    public void run() {
        for (int i = 1; i <= 5; i++) {
            if (i == 1)
                Thread.yield();

            System.out.println("From Thread A : i = " + i);
        }
        System.out.println("Exit from A");
    }
}

class B extends Thread {
    public void run() {
        for (int j = 1; j <= 5; j++) {
            System.out.println("From Thread B : j = " + j);

            if (j == 3)
                break;
        }
    }
}

class C extends Thread {
    public void run() {
        for (int k = 1; k <= 5; k++) {
            System.out.println("From Thread C : k = " + k);

            if (k == 1) {
                try {
                    Thread.sleep(1000);
                } catch (Exception e) {
                }
            }
        }
        System.out.println("Exit from C");
    }
}

class ThreadMethods {
    public static void main(String args[]) {
        A threadA = new A();
        B threadB = new B();
        C threadC = new C();

        System.out.println("Start thread A");
        threadA.start();

        System.out.println("Start thread B");
        threadB.start();

        System.out.println("Start thread C");
        threadC.start();
```

```
        System.out.println("End of main thread");
    }
}
```

Output:

```
Start thread A
Start thread B
Start thread C
End of main thread

From Thread A : i = 1
From Thread B : j = 1
From Thread C : k = 1

From Thread B : j = 2
From Thread B : j = 3

From Thread A : i = 2
From Thread A : i = 3
From Thread A : i = 4
From Thread A : i = 5
Exit from A

From Thread C : k = 2
From Thread C : k = 3
From Thread C : k = 4
From Thread C : k = 5
Exit from C
```

Problem No - 08: **Write a Java program to use priority of thread.**

Source Code :

```java
class A extends Thread {
    public void run() {
        System.out.println("threadA started");
        for (int i = 1; i <= 4; i++) {
            System.out.println("From Thread A : i = " + i);
        }
        System.out.println("Exit from A");
    }
}

class B extends Thread {
    public void run() {
        System.out.println("threadB started");
        for (int j = 1; j <= 4; j++) {
            System.out.println("From Thread B : j = " + j);
        }
        System.out.println("Exit from B");
    }
}

class C extends Thread {
    public void run() {
        System.out.println("threadC started");
        for (int k = 1; k <= 4; k++) {
            System.out.println("From Thread C : k = " + k);
        }
        System.out.println("Exit from C");
    }
}

class ThreadPriority {
    public static void main(String[] args) {
        A threadA = new A();
        B threadB = new B();
        C threadC = new C();

        threadC.setPriority(Thread.MAX_PRIORITY);
        threadB.setPriority(threadA.getPriority() + 1);
        threadA.setPriority(Thread.MIN_PRIORITY);

        System.out.println("Start thread A");
        threadA.start();

        System.out.println("Start thread B");
        threadB.start();

        System.out.println("Start thread C");
        threadC.start();

        System.out.println("End of main thread");
    }
}
```

Output:

```
Start thread A
Start thread B
Start thread C
End of main thread

threadB started
From Thread B : j = 1
From Thread B : j = 2

threadA started
From Thread A : i = 1
From Thread A : i = 2
From Thread A : i = 3
From Thread A : i = 4
Exit from A

threadC started
From Thread C : k = 1

From Thread B : j = 3
From Thread B : j = 4
Exit from B

From Thread C : k = 2
From Thread C : k = 3
From Thread C : k = 4
Exit from C
```

Problem No - 09: **Write a client-server TCP socket program in java that the server listens for connection requests, and whatever message the client sends, the server converts it to uppercase and sends it back.**

Source Code (Server):

```java
import java.io.*;
import java.net.*;

public class TCPServer {
    public static void main(String[] args) throws Exception {
        ServerSocket ss = new ServerSocket(5000);
        System.out.println("Server is waiting...");
        Socket s = ss.accept();
        System.out.println("Client connected");
        BufferedReader br = new BufferedReader(
            new InputStreamReader(s.getInputStream()));
        PrintWriter pw = new PrintWriter(s.getOutputStream(), true);
        String msg = br.readLine();
        System.out.println("Received: " + msg);
        pw.println(msg.toUpperCase());
        s.close();
        ss.close();
    }
}
```

Source Code (Client):

```java
import java.io.*;
import java.net.*;

public class TCPClient {
    public static void main(String[] args) throws Exception {
        Socket s = new Socket("localhost", 5000);
        BufferedReader input = new BufferedReader(
            new InputStreamReader(System.in));
        PrintWriter pw = new PrintWriter(s.getOutputStream(), true);
        BufferedReader br = new BufferedReader(
            new InputStreamReader(s.getInputStream()));
        System.out.print("Enter message: ");
        String msg = input.readLine();
        pw.println(msg);

        String response = br.readLine();
        System.out.println("Server response: " + response);

        s.close();
    }
}
```

Output:

```
Server is waiting...
Client connected
Received: hello

Enter message: hello
Server response: HELLO
```

Problem No - 10: **Write a client-server UDP socket program in java that the server listens for connection requests, and whatever message within 1024 bytes the client sends, the server converts it to uppercase and sends it back after 6 ms.**

Source Code (Server):

```java
import java.net.*;

public class UDPServer {
    public static void main(String[] args) throws Exception {
        DatagramSocket ds = new DatagramSocket(5000);
        byte[] buffer = new byte[1024];

        System.out.println("UDP Server waiting...");

        DatagramPacket dp = new DatagramPacket(buffer, buffer.length);
        ds.receive(dp);

        String msg = new String(dp.getData(), 0, dp.getLength());
        System.out.println("Received: " + msg);

        Thread.sleep(6);

        String upper = msg.toUpperCase();
        byte[] sendData = upper.getBytes();

        DatagramPacket sendPacket = new DatagramPacket(
            sendData, sendData.length,
            dp.getAddress(), dp.getPort());

        ds.send(sendPacket);
        ds.close();
    }
}
```

Source Code (Client):

```java
import java.net.*;
import java.io.*;

public class UDPClient {
    public static void main(String[] args) throws Exception {
        DatagramSocket ds = new DatagramSocket();

        BufferedReader input = new BufferedReader(
            new InputStreamReader(System.in));

        System.out.print("Enter message: ");
        String msg = input.readLine();

        byte[] sendData = msg.getBytes();
        InetAddress ip = InetAddress.getByName("localhost");

        DatagramPacket dp = new DatagramPacket(
            sendData, sendData.length, ip, 5000);

        ds.send(dp);
```

```java
        byte[] buffer = new byte[1024];
        DatagramPacket receivePacket = new DatagramPacket(buffer,
           buffer.length);
        ds.receive(receivePacket);

        String response = new String(
            receivePacket.getData(), 0, receivePacket.getLength());

        System.out.println("Server response: " + response);

        ds.close();
    }
}
```

Output:

```
UDP Server waiting...
Received: java

Enter message: java
Server response: JAVA
```